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High-precision measurement of Cooper-pair mass using rotating spherical-shell superconductor

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ABSTRACT

A new method is proposed for determining the mass of a Cooper pair at extremely low temperatures by measuring the London moment of a rotating spherical-shell superconductor. The relationship between the magnetic field generated by the rotating superconductor and the angular speed of the latter is examined. The other sample parameters are optimized to maximize the precision of the measurement. Values of the Cooper-pair mass are obtained with several bulk type-I superconductors.

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1. Introduction

The generation of a magnetic field by a rotating macroscopic superconductor was described first by Becker et al. [1] and then by London [2]. The magnetic moment of this generated field, known as the London moment, has been shown to be proportional to the effective mass of a Cooper pair and the angular velocity. The phenomenon was later demonstrated experimentally by Hildebrandt [3], Bol and Fairbank [4], and King et al. [5]. The effective Cooper-pair mass (CPM) in the expression for the magnetic field induced by a rotating superconductor of a given geometry [6–8] is not exactly twice the mass m_e of a free electron. Several studies have demonstrated CPM correction [9–16], according to which the deviation of the CPM from $2m_e$ is due to the large velocity and kinetic energy that contribute to the total mass in special relativity. In addition, the moving internal electrostatic potential of the rotating lattice in the rotating superconductor adds an extra term to the total CPM [13]. Thus, the effective CPM in a superconductor is an essential parameter that indicates the fundamental physics of superconductivity with much information about the lattice structure and the electron–electron interaction in a superconductor.

To date, several experimental schemes have been used to determine the effective CPM. Tate et al. [17,18] measured it by means of a rotating superconducting ring and found the ratio of the effective mass to $2m_e$ to be 1.000084(21), which contradicted the theoretical value of 0.999992 without considering the contribution of the finite size of the superconducting samples. In fact, they neglected the London penetration depth (LPD) that appears as an important parameter in the London moment formula for a finite-size superconductor, being the distance that an external field can penetrate inside the superconductor and thus determining how the supercurrents or electrons are distributed in the sample. The Doppler splitting of plasmon resonance [19] and the Bernoulli effect [20] in a superconductor by Mishonov potentially contains a huge systematic error associated with the several complex apparatus. In addition, although measuring the London moment has been proposed in several scientific schemes, especially the Gravity Probe B project to test general relativity [21–24], the practical

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applications of the measurement in examining the lattice structure of superconductors through the CPM are yet to be explored thoroughly.

Herein, we propose a new, precise method for measuring the effective CPM at ultralow temperature (10^{-4} K) by measuring the London moment of a rotating spherical-shell superconductor via a highly sensitive superconducting quantum interference device (SQUID) magnetometer [25,26]. In the measurement, along with the relativistic factor mentioned in the previous schemes, we consider the contribution of the LPD, the size of the superconducting shell, and the inhomogeneity of induced magnetic field at varying distances can be considered.

We begin by revisiting the expression for the magnetic field in SI units along an axis parallel to the rotation axis but separated by radial distance r that is induced by a spherical-shell superconductor of outer radius a_o, inner radius a_i, and rotating with angular velocity ω, namely [6,7]

$$B = \frac{2(1+\varepsilon)m^*}{e^*} \times \frac{a_o^3}{r^3}\left(1+\frac{3}{\left(\frac{a_o}{\lambda_L}\right)^2}-\frac{3}{\frac{a_o}{\lambda_L}}\left(\frac{\left(3+\left(\frac{a_i}{\lambda_L}\right)^2\right)+3\frac{a_i}{\lambda_L}\tanh\left(\frac{a_o-a_i}{\lambda_L}\right)}{\left(3+\left(\frac{a_i}{\lambda_L}\right)^2\right)\tanh\left(\frac{a_o-a_i}{\lambda_L}\right)+3\frac{a_i}{\lambda_L}}\right)\right)\omega \tag{1}$$

where λ_L is the LPD, m^* is the effective CPM, $e^* = 2e$ is the charge of a Cooper pair. The hyperbolic term containing the LPD expresses the significant contribution from the distribution of supercurrent to the measurement precision. Also, the relativistic correction [10,11,13] is introduced in equation (1) as the appearance of Lorentz factor $(1+\varepsilon)$. In the proposed method, we will investigate the linear relationship between this magnetic field and the angular velocity of the superconducting shell and substantially compute the effective CPM m^* based on the measureable parameters, namely a_o, a_i, r, e, ε, and $\vartheta = B/\omega$:

$$m^* = \frac{\vartheta e^*}{2(1+\varepsilon)} \times \frac{r^3}{a_o^3}\left(1+\frac{3}{\left(\frac{a_o}{\lambda_L}\right)^2}-\frac{3}{\frac{a_o}{\lambda_L}}\left(\frac{\left(3+\left(\frac{a_i}{\lambda_L}\right)^2\right)+3\frac{a_i}{\lambda_L}\tanh\left(\frac{a_o-a_i}{\lambda_L}\right)}{\left(3+\left(\frac{a_i}{\lambda_L}\right)^2\right)\tanh\left(\frac{a_o-a_i}{\lambda_L}\right)+3\frac{a_i}{\lambda_L}}\right)\right)^{-1}.$$

The systematic uncertainty δ of the measurement is

$$\delta m^* = \left[\left(\frac{\partial m^*}{\partial\vartheta}\Delta\vartheta\right)^2+\left(\frac{\partial m^*}{\partial a_o}\Delta a_o\right)^2+\left(\frac{\partial m^*}{\partial a_i}\Delta a_i\right)^2+\left(\frac{\partial m^*}{\partial r}\Delta r\right)^2 +\left(\frac{\partial m^*}{\partial \lambda_L}\Delta\lambda_L\right)^2+\left(\frac{\partial m^*}{\partial e}\Delta e\right)^2+\left(\frac{\partial m^*}{\partial \varepsilon}\Delta\varepsilon\right)^2\right]^{1/2}, \tag{2}$$

and the relative error is

$$\varepsilon_r = \frac{\delta m^*}{m^*}, \tag{3}$$

where the values of the "Δ" terms come from the error budget of the experimental system (Table 1). The calculation of uncertainty ϑ is related to magnetic field and angular velocity errors and given by least square fitting method [27]while the LPD is measured as in previous studies [28,29]. The experimental setup and procedure are described in the Supplementary Material.

Fig. 1 shows the experimental linear relationship between the magnetic field and the angular velocity. We conducted the same procedure on different type-I superconductors to obtained the CPM in different lattice structures. Besides, Fig. 2a–c illustrate how the induced magnetic field of the type-I superconductors depends on the radial distancer, the size of the shell, and the thickness of the superconducting layer respectively. The field amplitude decreases with the radial distance but increases with the radius' quartz sphere. Considering the thickness dependence, the magnetic field amplitude increases with the shell thickness and saturates at a specific thickness that depends on the superconducting material. The findings also imply that the formula of induced magnetic field (1) at zero-thickness limitation $\Delta a = 0$ coincides to the original London moment $B = -\frac{2m^*}{e^*}\omega$ i.e neglecting the finite size is only appropriate for thin superconducting shell. The results of the mass measurements conducted on the type-I superconductors are shown in Fig. 3, and Table 1 lists the factors that contribute to the measurement error. Despite the differences in actual value of relativistic correction, both microscopic [9–11] and gravitational thermoelectric [12,14] as well as thermodynamics theories [15,16] agree that this correction is extremely small and thus this effect only presents as the error budget $\Delta\varepsilon$ depending on the superconducting sample. The uncertainty of measurement appears to be extremely smaller than the measured values (average error of 2.5 ppm), which demonstrates the precision of measurement compared with the previous schemes [17,18] (21 ppm). The uncertainty in the angular velocity arises not only from averaging the velocities of the incoming and outgoing gas jets but also from the non-unified velocity distribution of the four gyroscopes. Moreover, there is a 0.001 ppm error coming from the polishing process which makes gyroscopes imperfectly spherical. The uncertainty associated with electrostatic charging of the rotor surface does not appear in the present experiment as in previous CPM measurements using a superconducting ring-spun up using a rotor [17,18].

Table 1
Systematic errors in experimental apparatus. (*: The error values arise from relativistic correction11 $\Delta\varepsilon$ belongs to Aluminum (Al)).

Error budget	
Radius of quartz core	±0.05 cm
Thickness of superconducting layer	±0.05 μm
Radial distance to sensor	±0.05 cm
Magnetic field	0.5 ppm
Angular velocity	±0.3 rad/s
Ratio of magnetic field to angular velocity	±0.5%
Relativistic correction	65 ppm (Al)*
Electron charge	0.5 ppm
$e = 1.602176462 \times 10^{-19}$C	
Electron mass	0.5 ppm
$m_e = 9.10938188 \times 10^{-31}$kg	

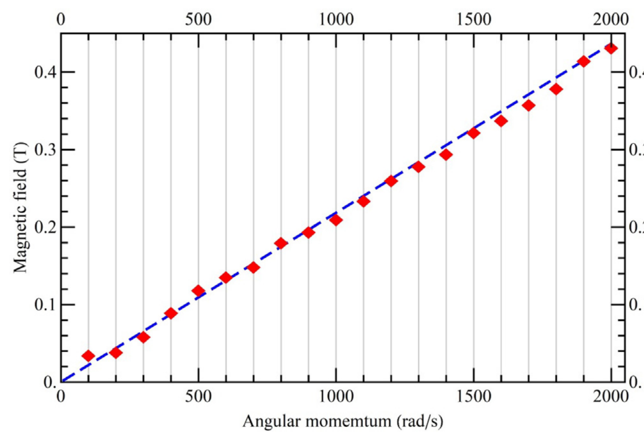

Fig. 1. Dependence of induced magnetic field outside rotating spherical-shell superconductor Sn on angular velocity, with a 2 μm layer of Sn, a fused-quartz core of 2 cm radius, and a radial distance to the SQUID sensor of 5 cm at 10^{-4} K.

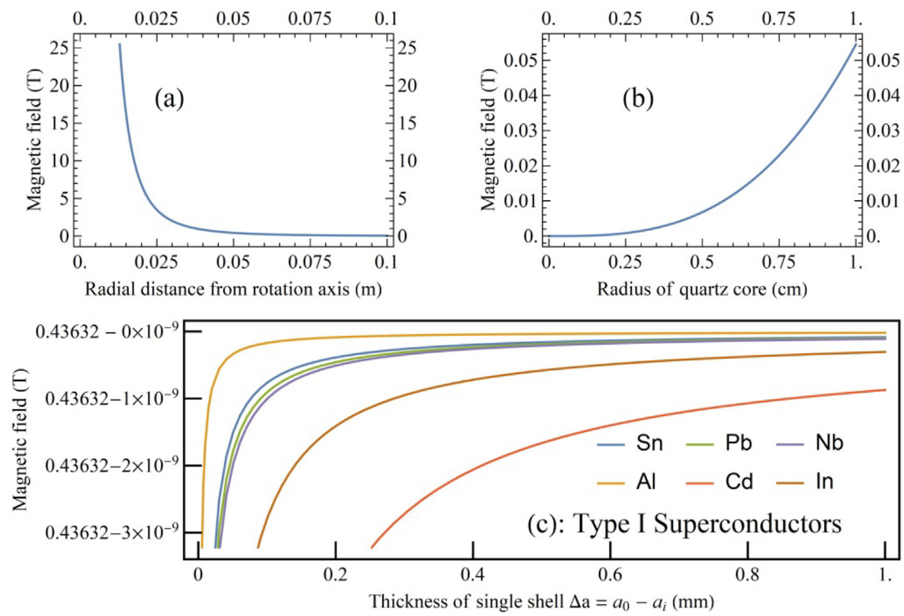

Fig. 2. Dependence of magnetic field inducing by a spherical-shell superconductor rotating with a velocity of 2000 rad/s at 10^{-4} K: (a) on radial distance from Helmholtz coil assembly to rotation axis when fused-quartz core radius is 2 cm; (b) on size of fused-quartz core when the radial distance is 5 cm with for 2 µm layer of Sn and (c) on the thickness in the case of a single type-I superconducting shell when fused-quartz core radius is 2 cm while the radial distance is 5 cm.

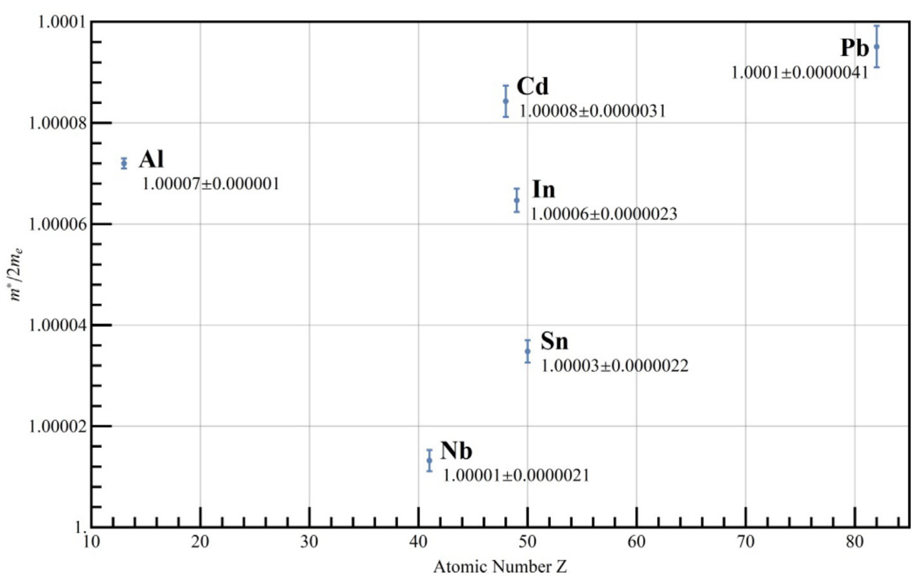

Fig. 3. Result of mass measurement in type-I superconductors.

In conclusion, we have proposed a method to precisely obtain the CPM based on a rotating superconducting spherical shell at ultralow temperature considering the finite size of the superconducting sample as well as involving the LPD in calculating the induced magnetic field. The measurement was carried out on different superconducting materials, and the error budget from different sources was provided. Based on this error budget, the high precision of measurement is guaranteed with significantly smaller error (average error of 2.5 ppm) than the previous one [17,18] (21 ppm). The proposed method is more precise than other methods proposed previously because it considers (i) the distribution of supercurrent in the superconducting sample for the specific geometry and (ii) the inhomogeneity of the magnetic field generated by the sample at different distances.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.matlet.2019.127176.

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Corrigendum

Corrigendum to “High-precision measurement of Cooper-pair mass using rotating spherical-shell superconductor” [Mater. Lett. 262 (2020) 127176]

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